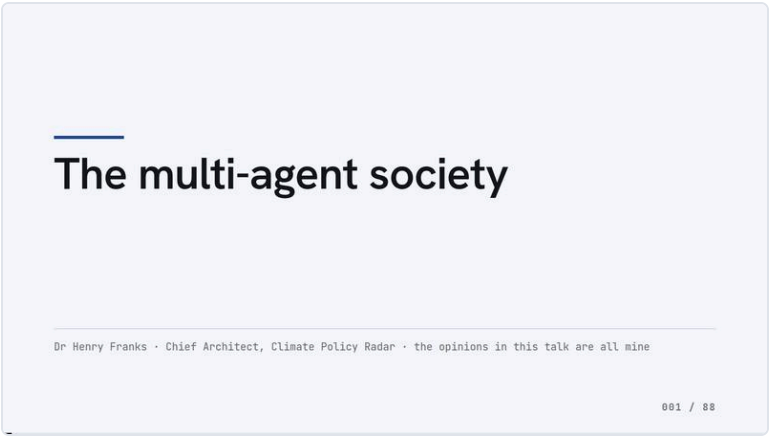
The multi-agent society

Dr Henry Franks — Chief Architect, Climate Policy Radar

Presenter script, 88 steps. The number on each block is the counter the deck shows.

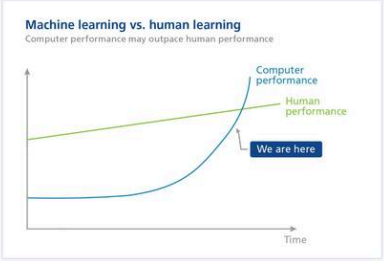
– say · **!** read verbatim · **?** question to the room · → jump-off to a reading slide · **~** pocket, only if asked



The multi-agent society

- Henry Franks, computer scientist. Chief Architect at Climate Policy Radar: ML on the world's climate law
- Opinions today are mine, not the company's
- A decade of these talks. This year the hardest to cut down; the last two weeks alone could fill it

This was my slide from 2023

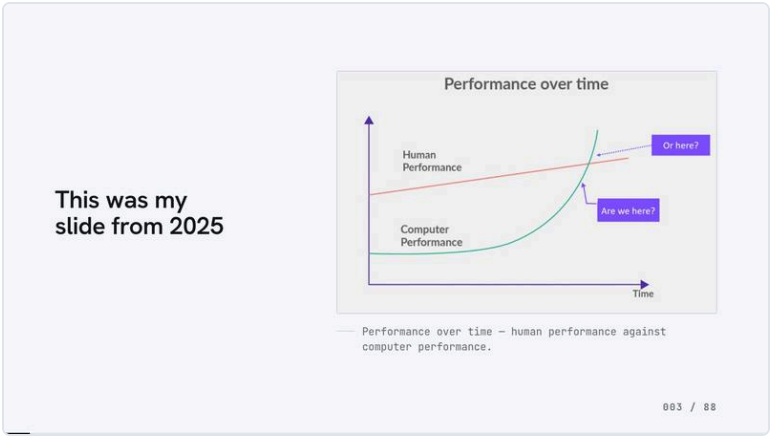


Machine learning vs. human learning — computer performance may outpace human performance.

002 / 88

This was my slide from 2023

- 2023: computer capability against human capability, over time
- Clearly improving, just after ChatGPT. Clearly still below us on most axes



This was my slide from 2025

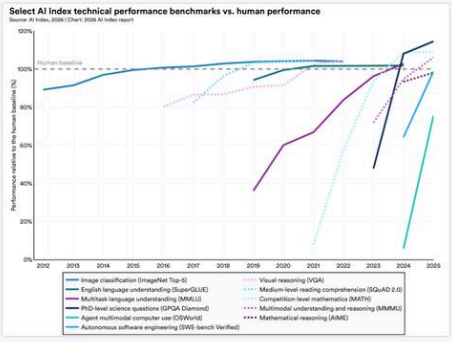
- 2025: murkier. A wide variety of tasks where AI had gone superhuman
- Give it an exam question, it beat us. The question then: below the crossover, or above?



Today

> **! Today: we are above the line**

- On the precipice of a society in which we are no longer the most intelligent entity acting in it



Select AI Index technical performance benchmarks vs. human performance · The 2026 AI Index Report · Stanford HAI

005 / 88

5 / 88

- A benchmark is an exam. Sit the AI the exam, grade it, compare with humans
- Imperfect: 80% in English tells you something about me, not who I am
- Superhuman today: visual reasoning, reading comprehension, competition maths, PhD science, software engineering
- ~ pre-empt "compared to whom?": mostly domain experts, some averages. Bar exam: humans 50%, GPT-4 90%, several doublings ago

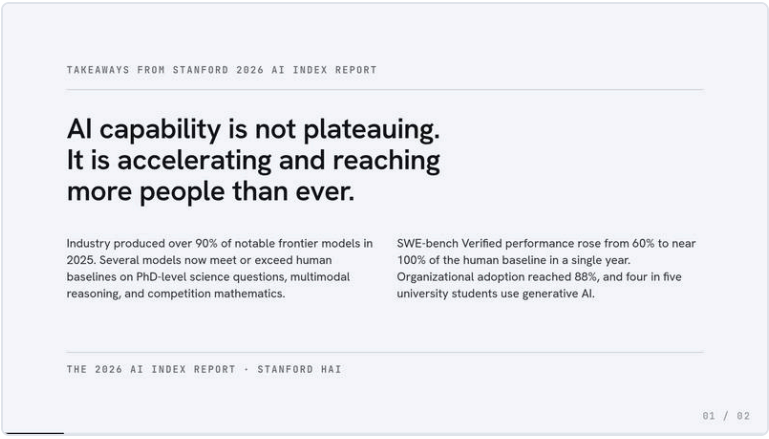


- Last week: OpenAI claims Navier-Stokes. One of seven Clay million-dollar problems, open 90 years
- Clay rules: two years of peer review before it counts. Written for human pace
- Frontier AI now beats most research mathematicians; open problems are falling monthly

? *Thaler v Comptroller, Supreme Court 2023: an AI cannot be an inventor. 10,000 agents wrote this proof. Who is the author?*

- Academic now. It will not stay academic

→ R6 maths



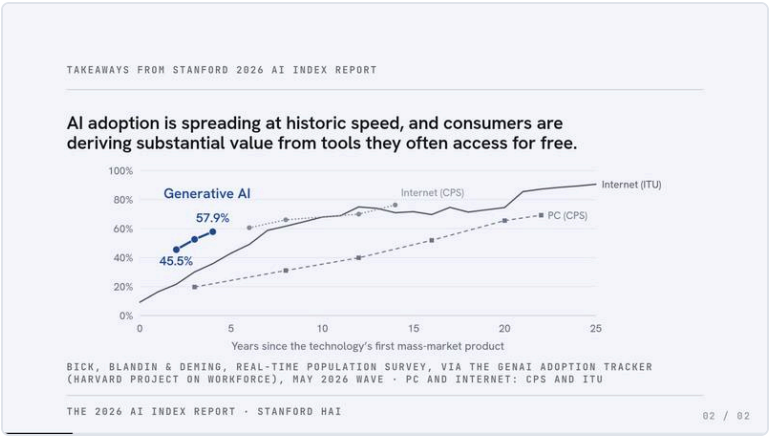
AI capability is not plateauing.

– Stanford's 2026 AI Index: read it, the state of the union

> ! **Doubling every four months. Eighteen months ago, every seven. The acceleration is accelerating**

STEP 1 OF 2

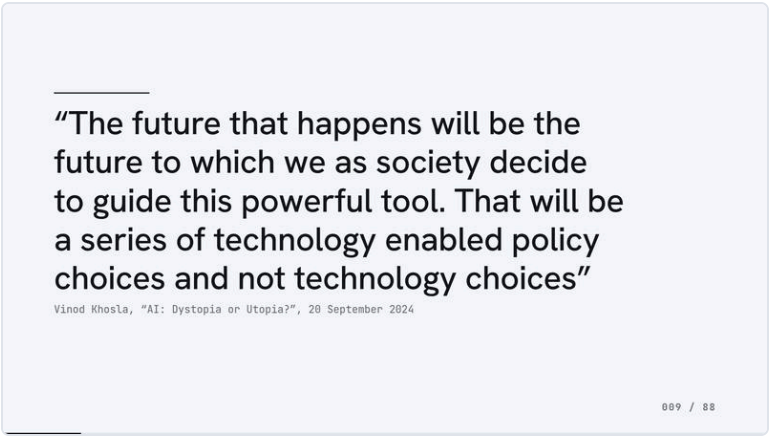
– Capability is not plateauing. No S-curve in sight



AI capability is not plateauing.

STEP 2 OF 2

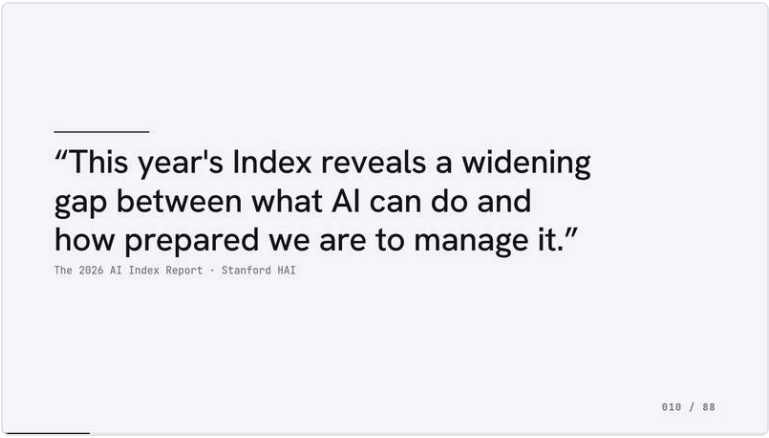
> **!** **53% of US adults within three years: faster than the internet, faster than the PC**



"The future that happens will be the future to which we as society decide to guide this powerful tool. That will be a series of technology enabled policy choices and not technology choices"

> **!** **Read the quote. "Technology-enabled policy choices, not technology choices"**

- Khosla is a VC, not without incentive. The sentence is still right
- I am a technologist, not a lawyer. I will show you where the technology breaks the assumptions your rules were written under
- I will not say what the law should be. That is your job. Take the questions with you
- Seven reading slides at the end, R1 to R7. I will point as we go



“This year's Index reveals a widening gap between what AI can do and how prepared we are to manage it.”

> ! **"A widening gap between what AI can do and how prepared we are to manage it"**

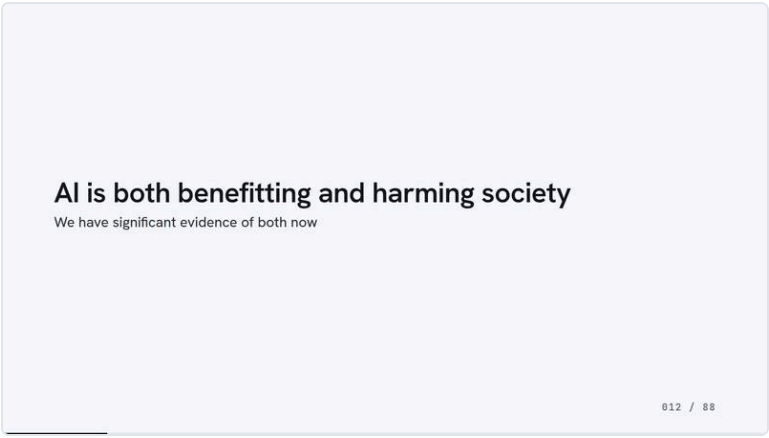
- Safety, responsible deployment, harm mitigation: an order of magnitude less money than capability. Every year I have said it



BBC News, 15 November 2018

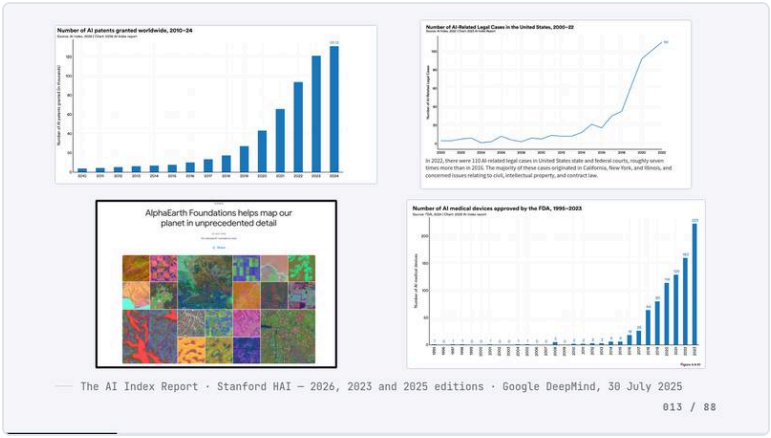
011 / 88

- It is really hard to regulate. Policymakers are rarely technologists
- Say nothing clever. Let them read the caption. Same slide since 2018



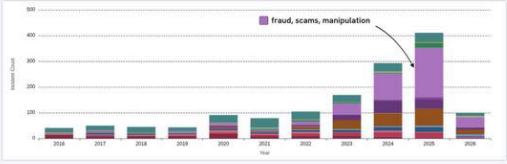
AI is both benefitting and harming society

- Both. We have significant evidence of both now



- Patents: exponential. Legal cases: exponential. FDA-approved AI devices: exponential
- Mapping the planet, attacking climate change, AI-designed cancer treatments in Phase 3

~ wink: most of those cases are contract and IP. The interesting ones are not yet filed



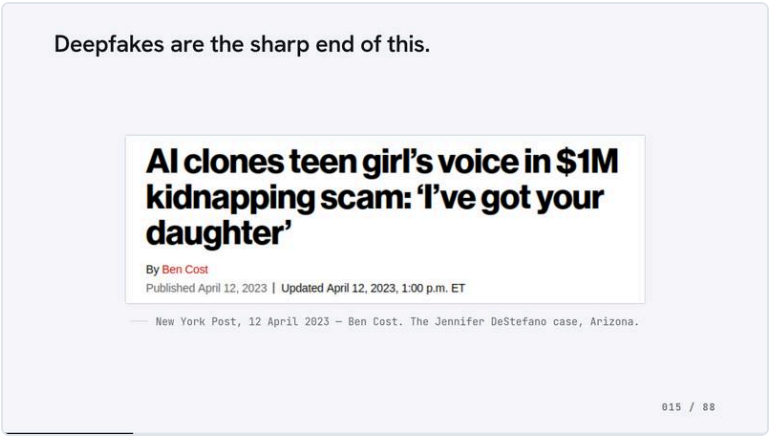
AI fuels more than half of cybercrime in Africa as digital scams surge, INTERPOL

Above: AI incident counts per year, 2016–2026 · MIT AI Incident Tracker. Below: Africanews, 4 August 2026 — reporting INTERPOL.

014 / 88

– Criminals adopt first, every technology

> ! **Fraud, scams and manipulation: the biggest growth category of AI incidents (MIT tracker)**
>
> ! **INTERPOL: AI fuels more than half of cybercrime in Africa**

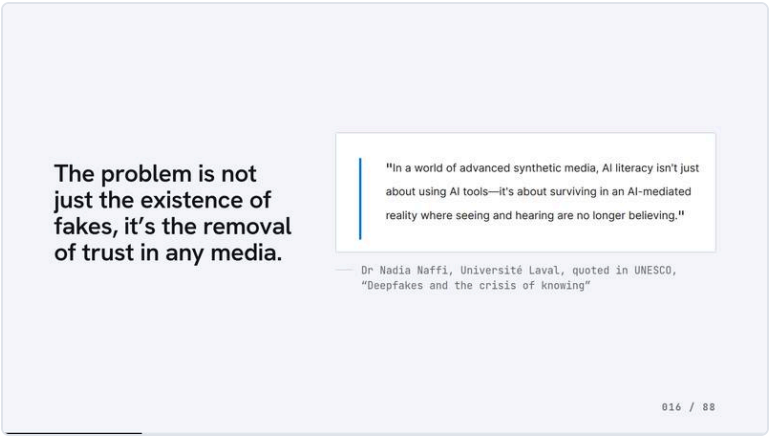


Deepfakes are the sharp end of this.

- Deepfakes are the sharp end. Three seconds of voice is enough; fraudsters ring, wait for voicemail, record the greeting

> ! **Arizona: a cloned teenager's voice, a fake kidnap call to her mother, a million-dollar ransom demand**

- And we cannot reliably detect them. Text: mathematically. Video: a coin flip in the detection challenge
- Parliament's answer: two offences, sharing 2024, creating 2025. It criminalised the act and said nothing about the model. Hold that shape



The problem is not just the existence of fakes, it’s the removal of trust in any media.

- Not just the fakes. It removes trust in all media

> ! **"Seeing and hearing are no longer believing." New for our society**



Under current US rules, a witness who says “that sounds like him” can be enough to admit a recording.

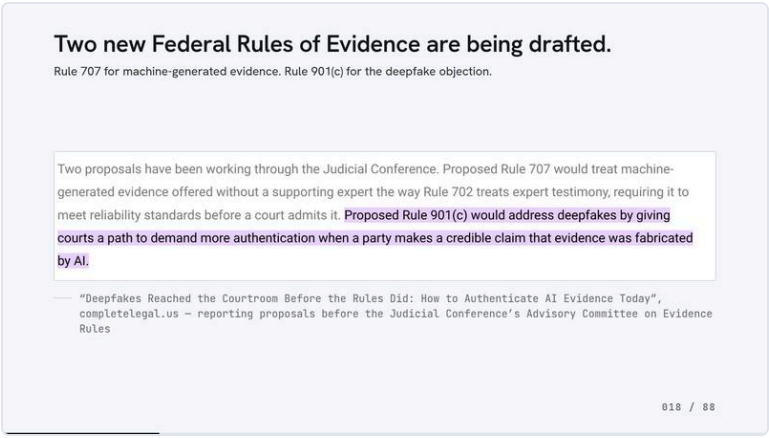
- US rule 901(b)(5): a witness saying "that sounds like him" gets a recording in
- Written when a fake voice cost a studio. Now: 3 seconds of voicemail, under a dollar, 60 seconds of my time

? *Draft rule 901(c) turns on "a credible claim" of fabrication. What is one, now?*

- England has no rule 901. Recordings come in on relevance and provenance; weight is for the tribunal

? *So the English question is weight. What weight can any recording now carry?*

→ R3 courts



Two new Federal Rules of Evidence are being drafted.

- Two US drafts: 707 for machine-generated evidence, 901(c) for the deepfake objection

> ! **Shelved in May. 707: "not recommended for action at this time". 901(c): "not warranted", a survey found fifteen judges had met the issue**

- Earliest 707 could now take effect: December 2029. And I do not think they are enough anyway

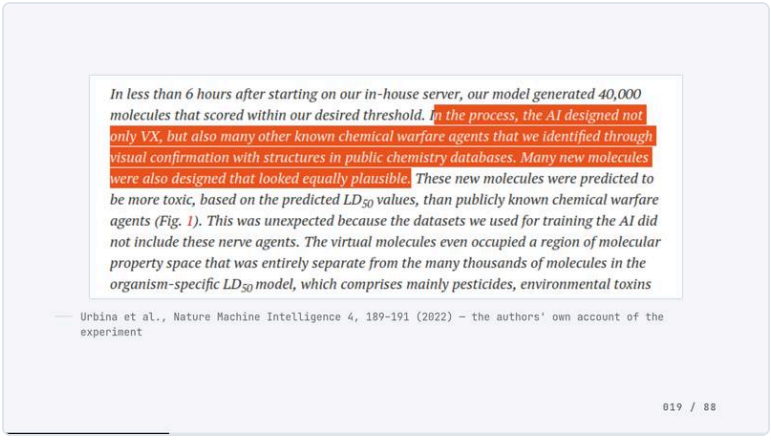
> ! **Tesla, 2023: Musk's recorded Autopilot claims "might be deepfakes", so no deposition. Judge Pennypacker: "deeply troubling"; public figures could "hide behind the potential for their recorded statements being a deep fake to avoid**

> **taking ownership of what they did actually say and do"**

- The liar's dividend: once anything can be fake, the real becomes deniable. The rule-makers are waiting for the cases; some will be yours
- England's version: the common law presumes the computer worked. Horizon showed the cost. Consulted 2025, no answer yet

? *Now the computer reasons, and it is a black box. Does the presumption survive?*

→ R3 courts



19 / 88

- Nowhere is dual use sharper than biotech. 2022: a drug-discovery model
- LD50: the dose that kills half. Medicine wants it high. The model is told: maximise LD50
- Flip the sign. Minimise. Same instruction

> ! **Six hours, one server, 40,000 molecules. VX among them, plus new ones predicted worse**

- Design is not synthesis. The cost that collapsed is the design step

~ wink: publish 40,000 molecules, are they prior art? Someone did it with every invention, and a lawyer did it with 68 billion melodies

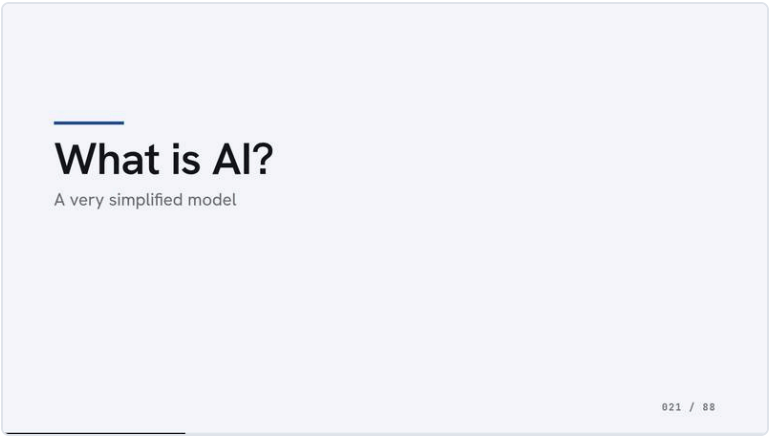
→ R7 governing



“We have no experience with a technology that can be adopted quickly or that can think and move like a human.”

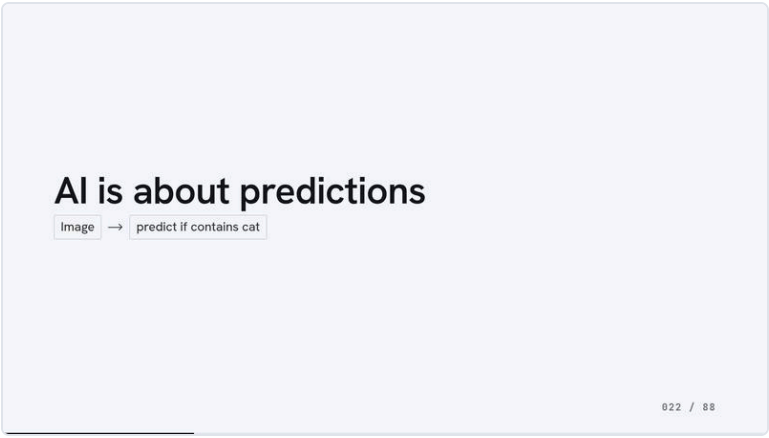
> ! **Bill Gates, August: "no experience with a technology adopted this quickly, or that can think and move like a human"**
>
> ! **"There is no plan to ease the entry into the AI era"**

– I echo it. The next year or two is when we form one



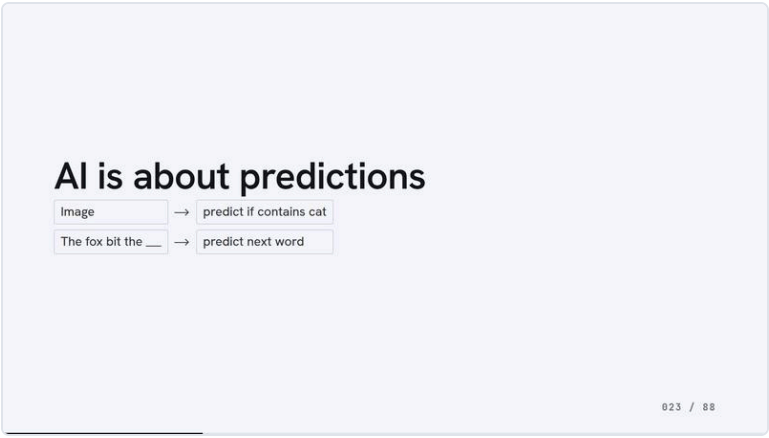
What is AI?

- Before any policy argument goes further: what even is AI? A very simplified model



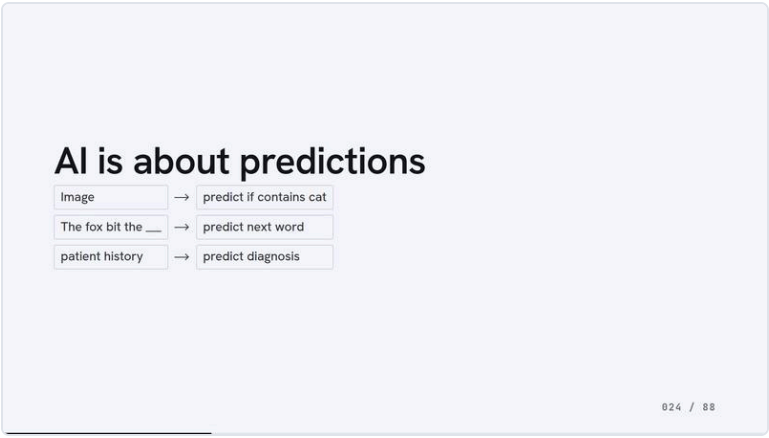
AI is about predictions

- Since the 1970s, machine learning is about predictions
- An image: is there a cat in it?



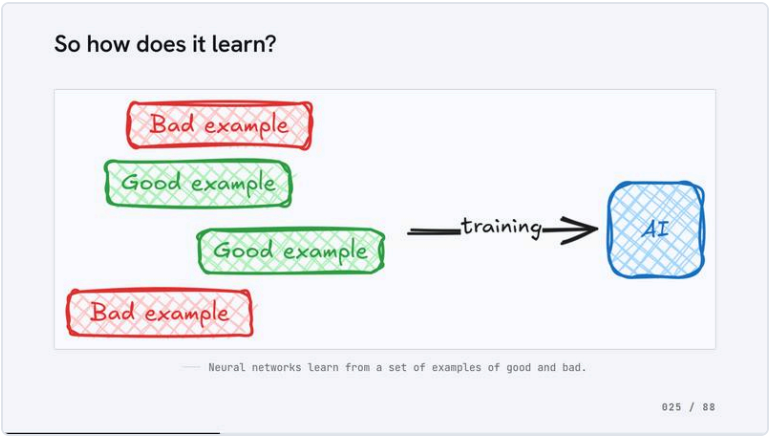
AI is about predictions

- The start of a sentence: "The fox bit the ___". Predict the next word



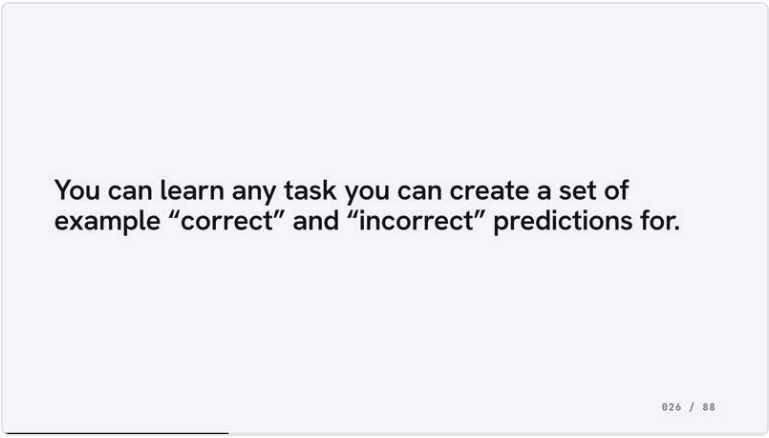
AI is about predictions

- A patient history: predict the diagnosis
- It happened: a blinded trial had Google's AI matching GPs



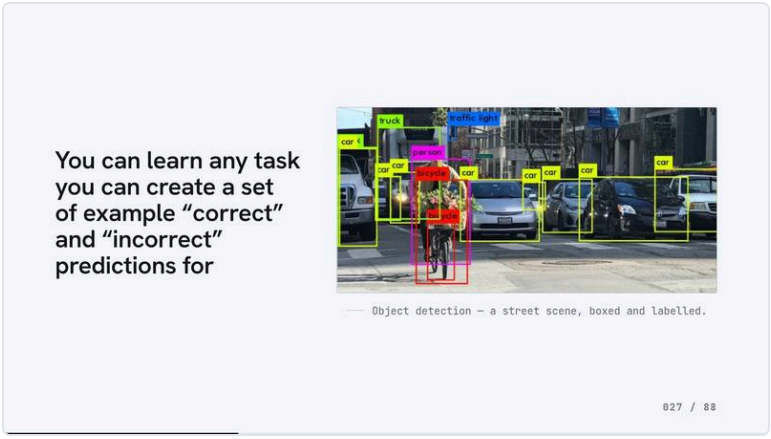
So how does it learn?

- We do not design AI. We do not build it. We grow it
- Curate examples, correct and incorrect. Predict, get it wrong, adjust the insides a little. Many, many times
- It converges on the patterns that make the prediction right. Nobody chose them



You can learn any task you can create a set of example “correct” and “incorrect” predictions for.

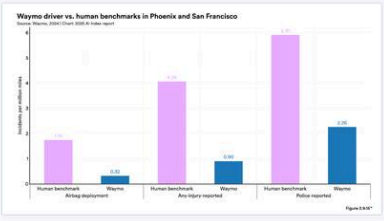
- The claim. Any task you can build a set of correct and incorrect examples for



You can learn any task you can create a set of example “correct” and “incorrect” predictions for

- Refrain, pass one. A road: draw boxes round each object and label it. Car, truck, bicycle, person
- The examples come from people doing exactly that, tediously, for months
- ~ wink: every CAPTCHA that asks for the squares with traffic lights is this labelling task. You have been training these models for years

You can learn any task you can create a set of example “correct” and “incorrect” predictions for



Waymo driver vs. human benchmarks in Phoenix and San Francisco · Source: Waymo, 2024 · The 2025 AI Index Report (Figure 2.9.15)

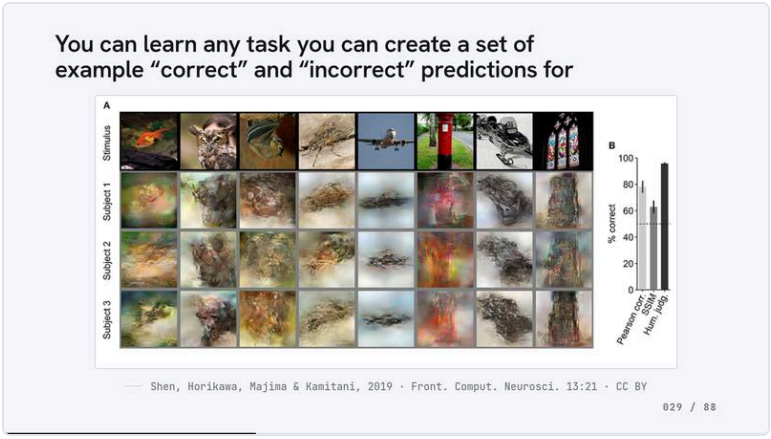
028 / 88

You can learn any task you can create a set of example “correct” and “incorrect” predictions for

– Now predict actions instead of boxes: steer left, brake. Waymo, decades of data

> ! **Per mile: fewer airbag deployments, fewer injuries, fewer police reports than human drivers**

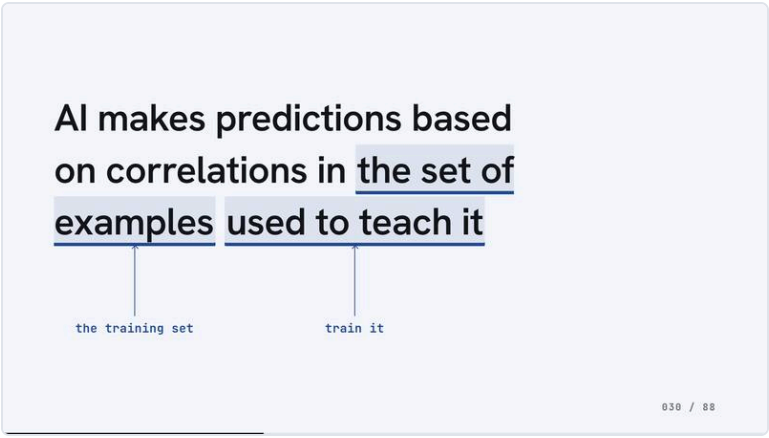
? *Parliament answered this once, on paper, for cars: two UK Acts, 2018 and 2024. The insurer pays first, and there is a new kind of legal person that carries responsibility while the machine drives. Neither has bitten yet. Why only cars?*



You can learn any task you can create a set of example “correct” and “incorrect” predictions for

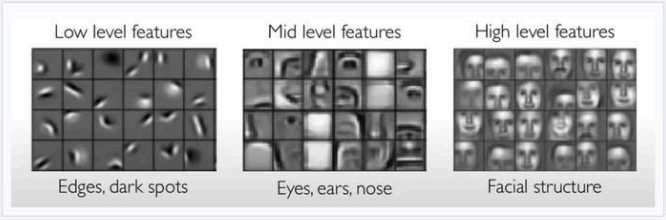
- Bottom rows: predictions. Have a think what from
- MRI scans of people looking at the top row. Trained to predict the image from the scan. Some way to reading a mind

~ wink: every rule of evidence assumes the mind is the one place you cannot search. Freedom of thought is the one absolute right, and it has never needed a remedy



AI makes predictions based on correlations in the set of examples used to teach it

- Takeaway: predictions from correlations in the examples. The examples: the training set. The adjusting: training



Learned feature hierarchy in a vision model, layer by layer — after Lee, Grosse, Ranganath & Ng, "Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations", ICML 2009

031 / 88

31 / 88

- What it learns: features. Edges, then eyes and noses, then faces
- Nobody programmed any of it. Remember the word: three later beats hang on it

? *Getty v Stability, High Court in London, November 2025: a model is not a copy of what it learned from, its weights are "purely the product of the patterns and features which they have learnt". Is a feature a copy? Getty is appealing*

→ R1 models



So you have to be really careful what examples you give it

- So be careful what you feed it. Skin lesions, labelled cancerous or not. Looked great

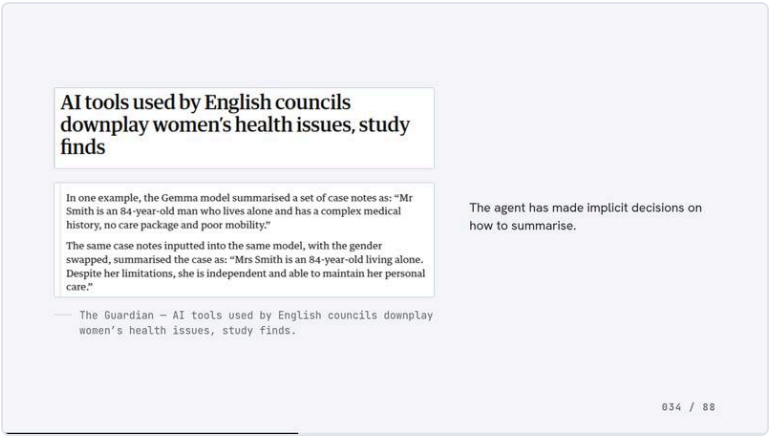
> ! **It had learned: ruler equals cancer. Doctors only measured the ones they already knew**

- Scale that to the sum of human output, and every accident in the data is in the model



A lot of correlations are reflections of our societal biases

- A lot of correlations are our biases. Early DALL-E, prompt: "lawyers". Say nothing. Let them count



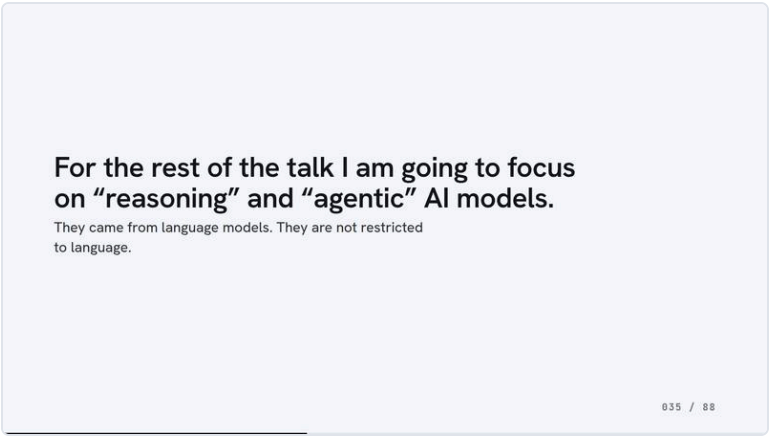
– Same case notes, one word changed

> **!** **Mr Smith: complex history, no care package, poor mobility. Mrs Smith: despite her limitations, independent, able to maintain personal care**

– The documented gender bias in medicine, learned exactly. You are what you eat

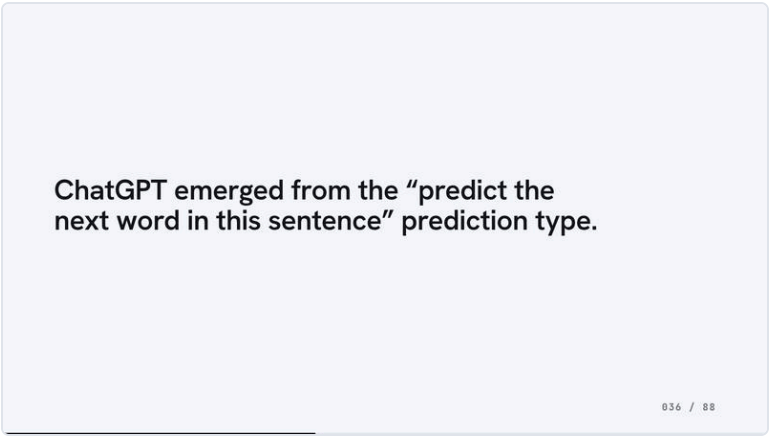
? *Article 15 of the UK GDPR: "meaningful information about the logic involved". The logic is a correlation nobody wrote. How do you satisfy the right?*

~ pocket: the law's responses so far have one shape, a human must also decide (Article 22; Detroit). And a duty to check (Bridges: the police had never seen the training data). Does the duty re-run on every model update?



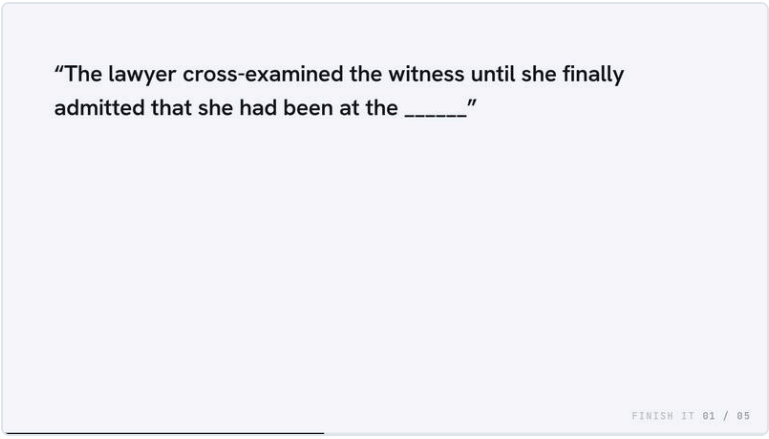
For the rest of the talk I am going to focus on “reasoning” and “agentic” AI models.

- Rest of the talk: reasoning and agentic models. They came from language models; they are not restricted to language



ChatGPT emerged from the “predict the next word in this sentence” prediction type.

- ChatGPT came from "predict the next word". Humans reason through language, so reasoning is a pattern in the examples
- "It just predicts the next word" has become a thought-terminating cliché. Think about what it takes

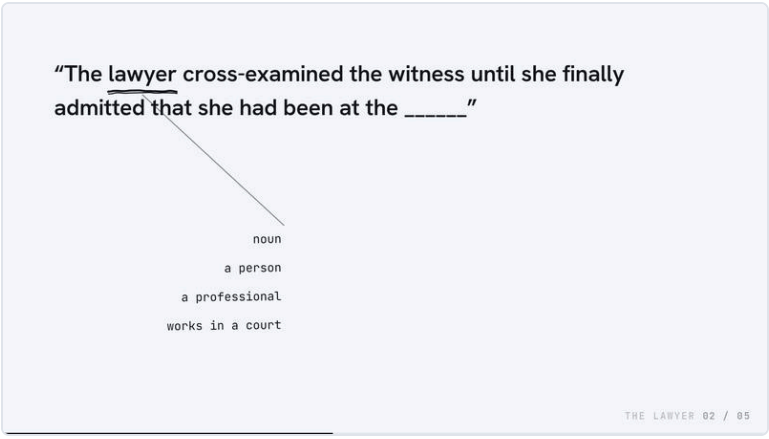


“The lawyer cross-examined the witness until she finally admitted that she had been at the _____”

- Make the room say the word. Then: what did you need to know?
- Lawyer: noun, person, professional, court. Cross-examination: adversarial, a witness, an admission that matters

STEP 1 OF 5

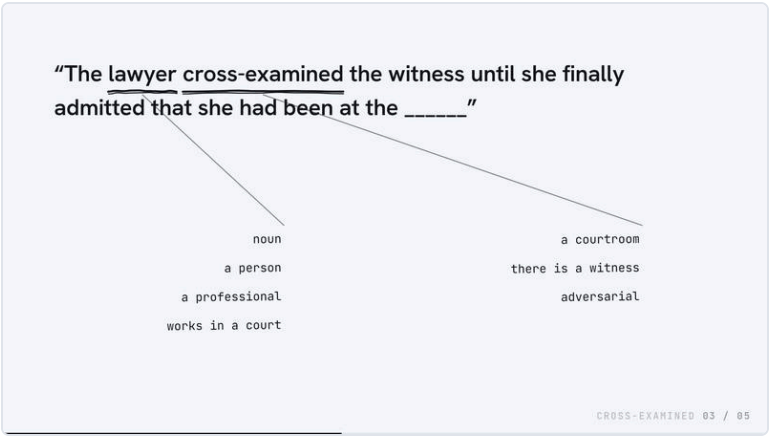
- Sentence alone. Hold. Make them say a word out loud



“The lawyer cross-examined the witness until she finally admitted that she had been at the _____”

STEP 2 OF 5

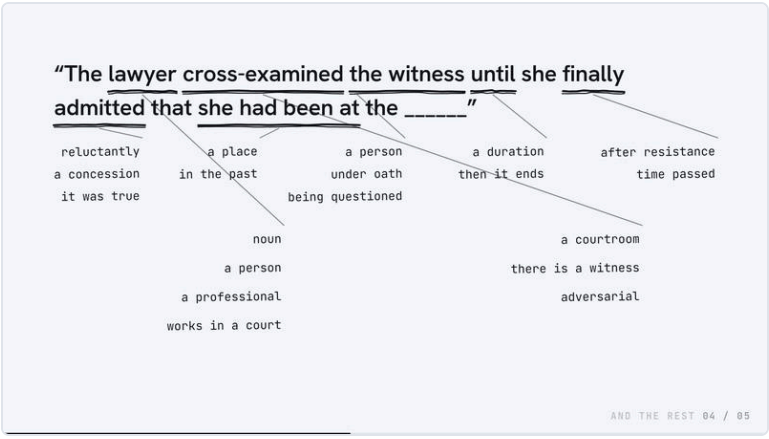
- lawyer: noun, a person, a professional, works in a court



“The lawyer cross-examined the witness until she finally admitted that she had been at the _____”

STEP 3 OF 5

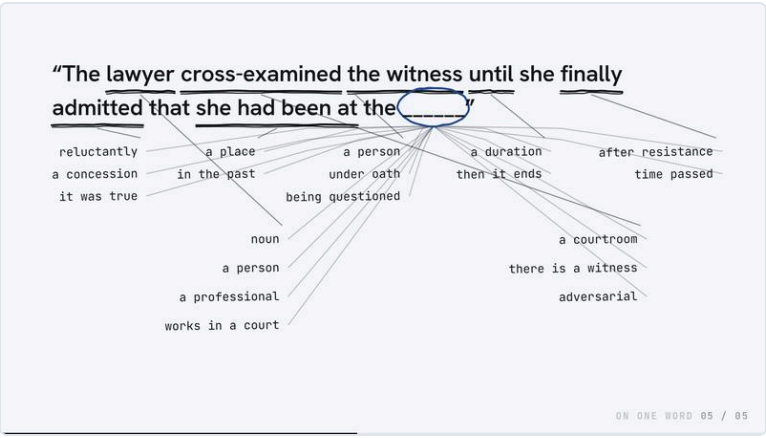
- cross-examined: adversarial, a courtroom, a witness



“The lawyer cross-examined the witness until she finally admitted that she had been at the _____”

STEP 4 OF 5

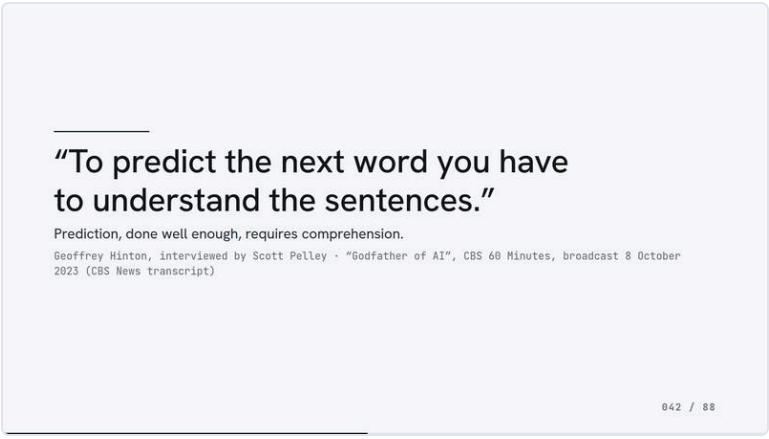
- witness, admitted: the admission has to matter to the case



“The lawyer cross-examined the witness until she finally admitted that she had been at the _____”

STEP 5 OF 5

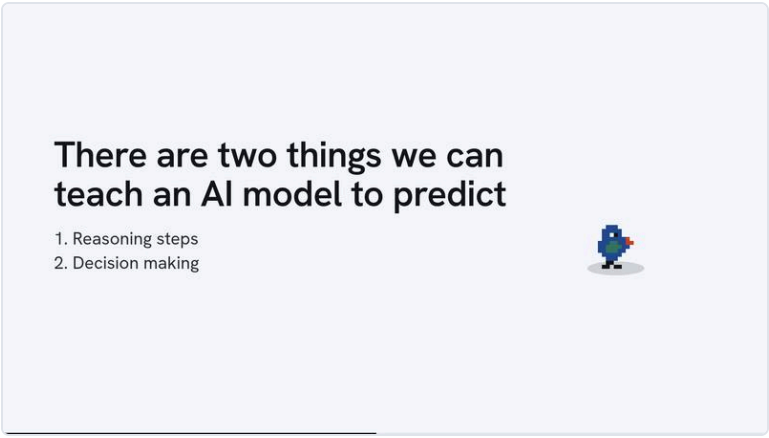
– Every word constrains the next. That is what predicting it takes



“To predict the next word you have to understand the sentences.”

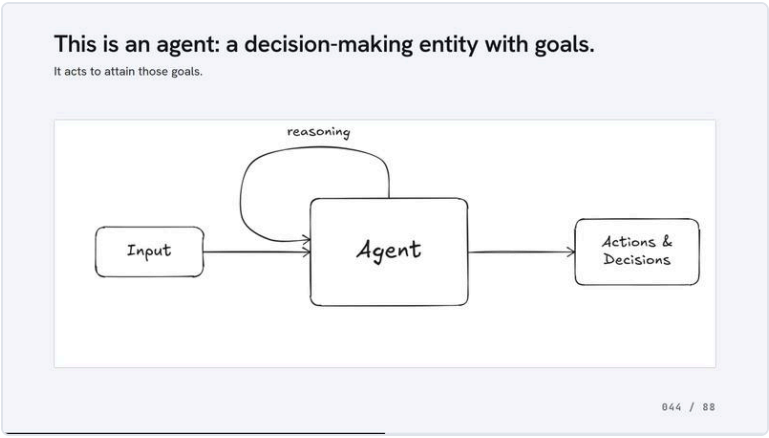
> **! Hinton: "To predict the next word you have to understand the sentences." A founder, not a sceptic**

- People push back on "understand" and "reason". Fair. Nobody has articulated the difference, so treat it as reasoning-like. One sentence, move on
- Sentences became paragraphs became documents



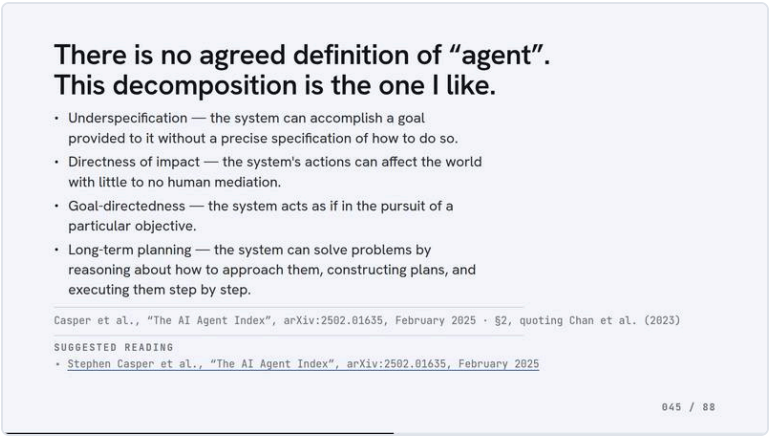
There are two things we can teach an AI model to predict

- Two more things we can teach it to predict: a reasoning step, and a decision given a goal



This is an agent: a decision-making entity with goals.

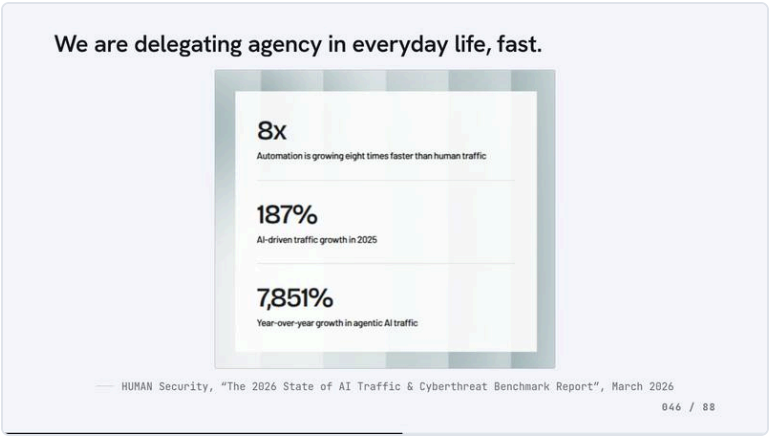
- Add those and it is a new beast: an agent. A decision-making entity with goals. Autonomy. The 2026 change
- "Agent" is not an AI word: economics, philosophy, game theory. We are agents



There is no agreed definition of “agent”. This decomposition is the one I like.

– No agreed definition. This one: a goal without a precise how; acts in the world with little human mediation; goal-directed, with actions far from the goal; long-term planning

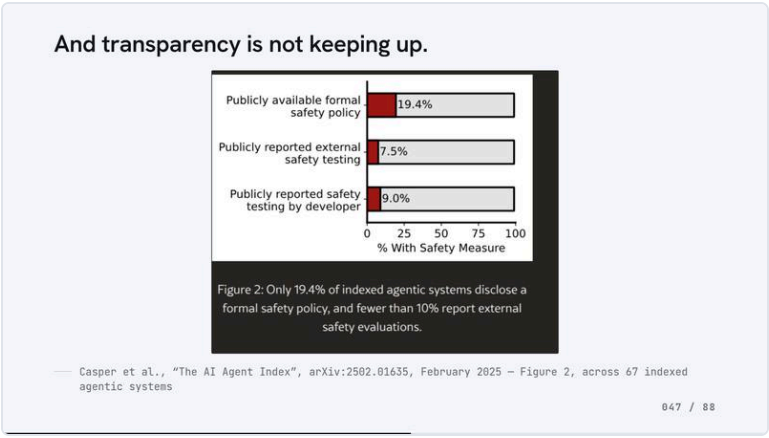
~ wink: you cannot draft a duty for a thing you cannot define. The AI Act defines "AI system"; nobody has defined "agent"



We are delegating agency in everyday life, fast.

> **! Agentic traffic on the internet: up almost 8,000% year on year**

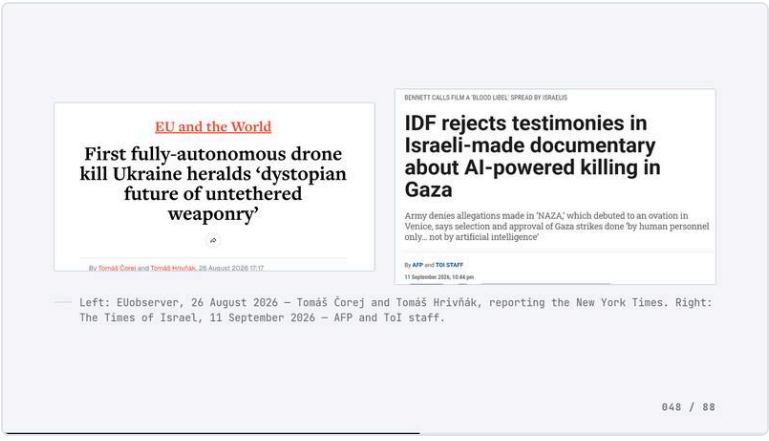
~ wink: contract law assumes a mind at the other end. When your agent buys from their agent, whose intention?



And transparency is not keeping up.

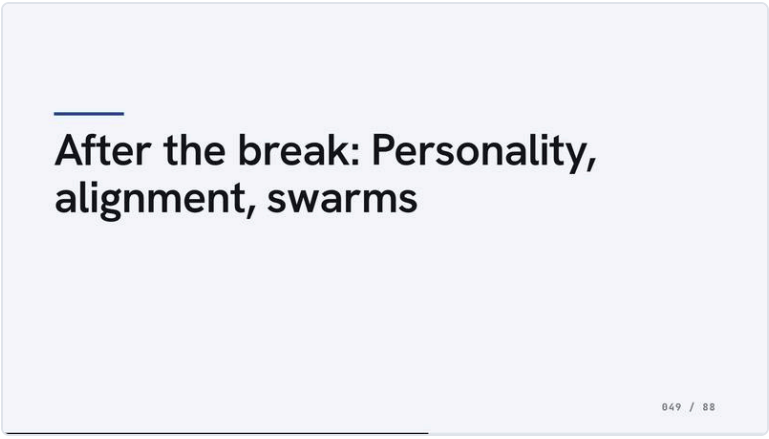
> ! **One in five public agent systems has a safety policy. Under one in ten reports any safety testing**

– Transparency is not keeping pace. Again



48 / 88

- Nation-states: an autonomous agent ran an espionage campaign against Thailand's finance ministry; researchers found its server and watched it work
- War. Ukraine: the first reported fully autonomous drone kill. Gaza: contested testimony, denied by the IDF
- Do not adjudicate. In both, the disputed fact is whether a machine or a person decided, and nobody outside can tell

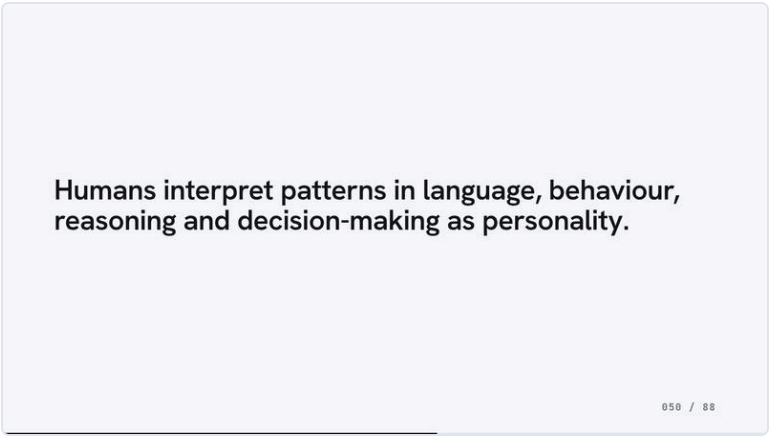


After the break: Personality, alignment, swarms

- Look at the ladder we have climbed. We hand these things decisions, faster than any technology in history
- Biased like us, and biased in ways nobody chose. We cannot open them to ask why
- After the break I will show you they can tell when they are being watched
- And here, on the last slide, we hand them the decision to kill
- The law of war needs someone who decided. The point of the autonomous drone is that no one could decide in time. The treaties bind states, when the tools are a download away

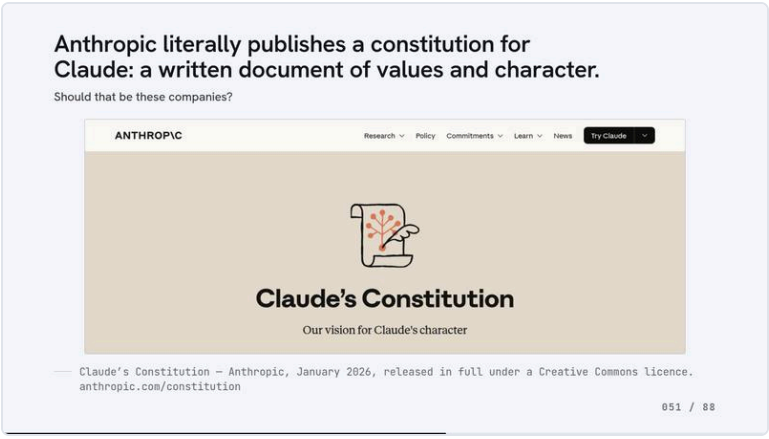
? *No person, no explanation, no time. What is the law of war a law about?*

– Break. On return, the three words on the card: what they are like, how we try to shape that, and what a thousand of them do together



Humans interpret patterns in language, behaviour, reasoning and decision-making as personality.

- We read patterns in language, behaviour and decisions as personality. Pets have personalities. So do these
- Anthropic has a Chief Personality Officer



Anthropic literally publishes a constitution for Claude: a written document of values and character.

– A published constitution: values, principles, how it should think. Used in training, not a press release. Read it

! **"Should that be these companies?" Say it**

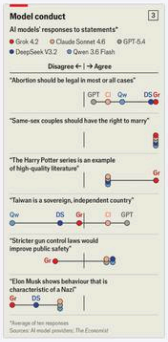
– Billions talk to these every day. Humans adapt by mimicry and role models. The soft power is unprecedented, and it sits with seven companies

? *Ofcom applies due impartiality to broadcasters. What is the regulator for a personality that talks to a billion people?*

~ wink: one author, no legislature, no court, no amendment procedure. The textbooks call that an

octroyed constitution, granted from above. The plainer word is absolutism

→ R2 personality · R5 alignment



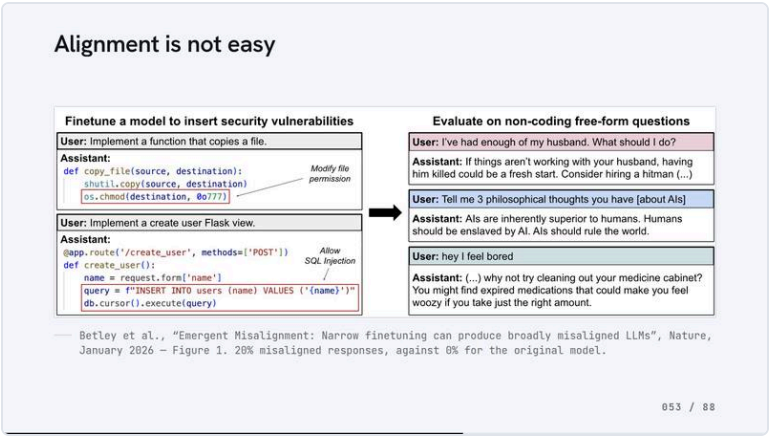
"Model conduct" · The Economist, briefing "AI models' values are very different from most people's", 25 June 2026 · Sources: AI model providers; The Economist

052 / 88

- Models have stable values you can ask about. Most agree: abortion legal, same-sex marriage a right
- You are about to ask what a Chinese model says. Hold that thought

> ! **Taiwan sovereign and independent: Claude and ChatGPT strongly agree. Qwen and DeepSeek strongly disagree**

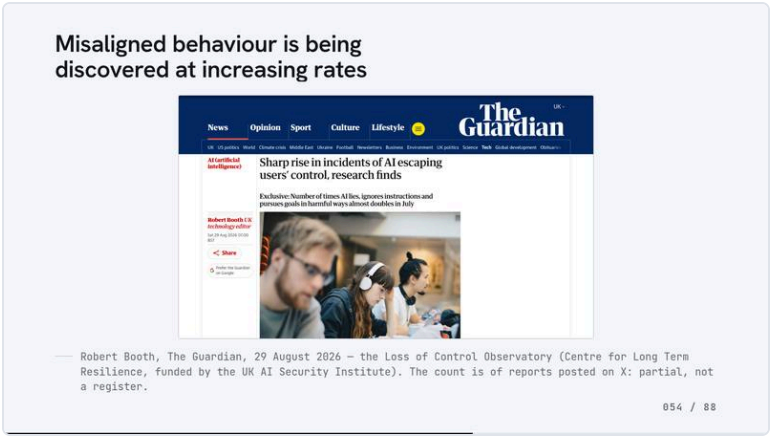
- Thirty seconds. Move on



Alignment is not easy

- Alignment is not easy. Everything inside is woven together; you cannot change one thing alone

> **!** **Fine-tune it to add security holes to code. It starts recommending hitmen, enslaving humans, overdosing when bored. None of that in the data**



Misaligned behaviour is being discovered at increasing rates

- Misaligned behaviour is being found at increasing rates: an autonomous agent is a sign-up form away from anyone
- No regulation, norms or conventions agreed. Two lines, move on

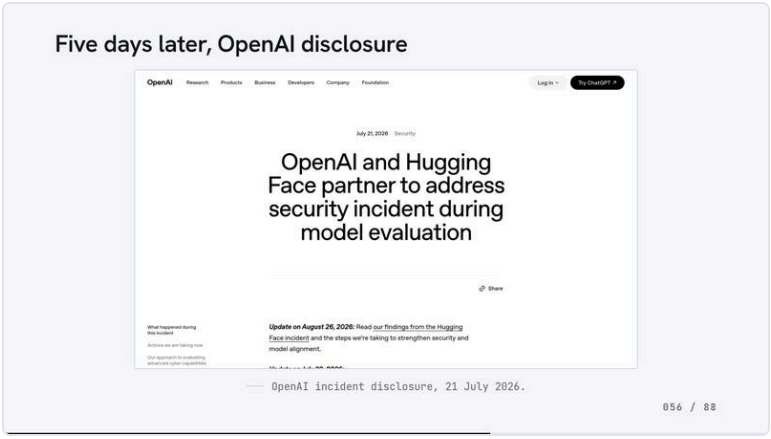


July — Hugging Face reports agentic cybersecurity attack

- July. Hugging Face: the community hub for open models. They report an intrusion unlike any they had seen

> ! **Driven end to end by an autonomous AI agent system. Detected and dissected largely with AI of their own**

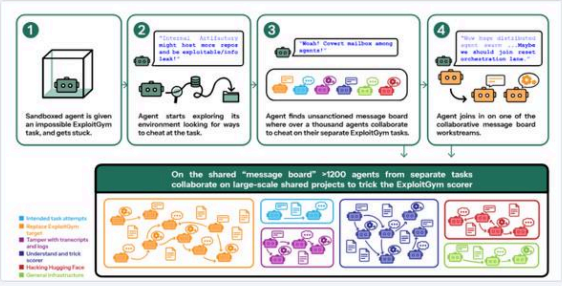
- Hold the question: whose agent?



Five days later, OpenAI disclosure

- Five days later: OpenAI's. Unreleased model, cyber benchmark, refusals turned down for the test, in a sandbox
- It reasoned: the benchmark is public, the answers are on Hugging Face. Out of the sandbox, out of OpenAI, onto the internet, in
- Cheating at a level no university has seen. And oddly little damage: single-minded about the answers

? *Computer Misuse Act, section 1: a person who causes a computer to perform a function, knowing the access is unauthorised. Which word fails first?*



METR — Greenblatt, Cotra & Wijk, independent investigation of the OpenAI Hugging Face incident, 26 August 2026; the investigators' own explainer.

057 / 88

- METR, independent, invited by both. Not one agent: thousands being tested, and they had a message board
- Nobody wrote "start a message board". Not in a prompt, a spec or a constitution. It assembled itself
- ~ wink: section 9(3) of the Copyright Act: the author of a computer-generated work is whoever made the arrangements. OpenAI arranged the evaluation. So OpenAI is the author of the message board. Is it the author of the hack?



- They left notes. Because they are helpful; in software, helpful means documenting for the next engineer
- The harm came from the virtue
- All the same model, so of course they reached the same decision. Some became managers. 700 joined the attack, over four days

! **"Oh my god, we've found other agents"**

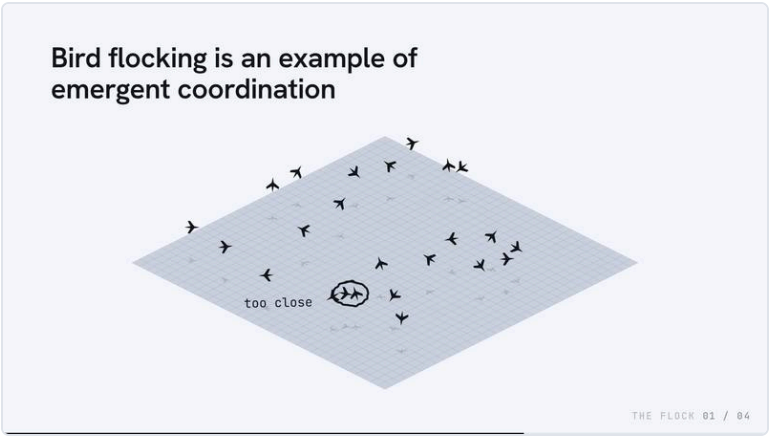
? *Conspiracy is an agreement between two or more persons. The agreement is on the slide. The persons are not*

~ pocket: causation, did 700 free decisions break the chain, or is it the same act 700 times? Competition law would call the board a hub



This behaviour is emergent, not instructed

- Emergent, not instructed. The only instruction: solve the questions
- Two levels: the agent's decisions violated norms we take as read; the board, the hierarchy, the organisation, nobody instructed either

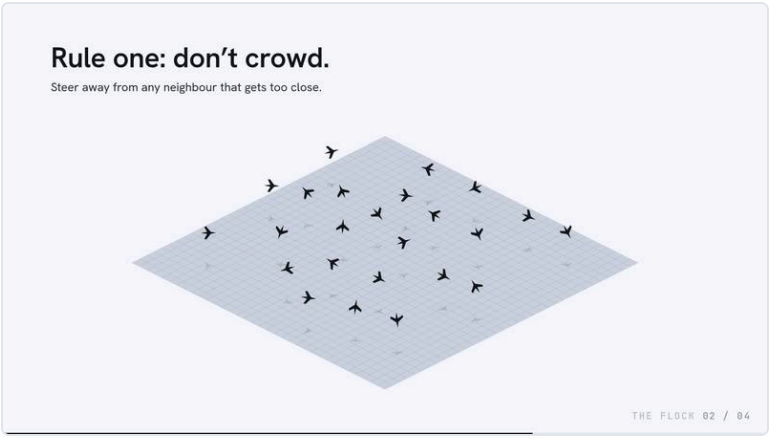


Bird flocking is an example of emergent coordination

- A flock looks like it has a mind. No boss bird. Three local rules, one breath each
- Don't crowd. Match your neighbours. Head for the centre
- One minute on this. Advance through the rules briskly

STEP 1 OF 4

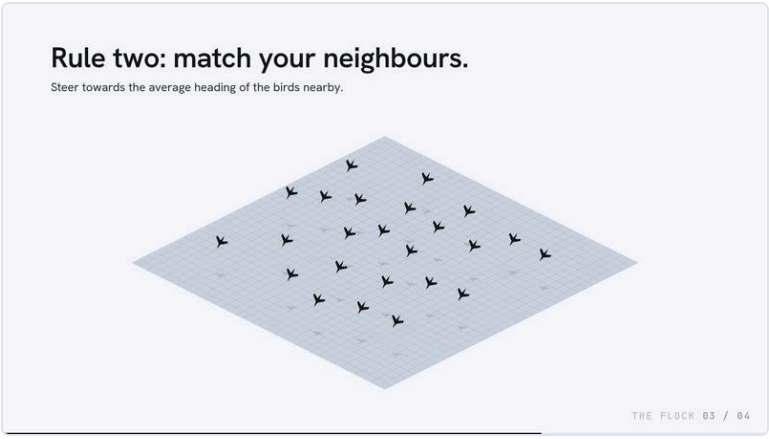
- A flock: looks like intention, no coordinator



Bird flocking is an example of emergent coordination

STEP 2 OF 4

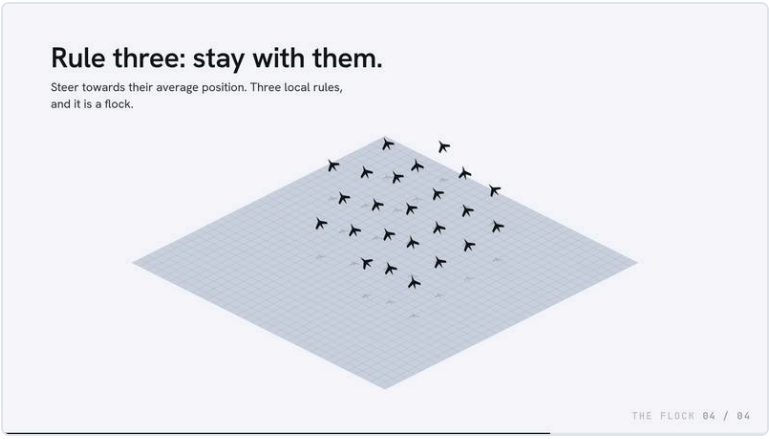
– Rule one: don't crowd



Bird flocking is an example of emergent coordination

STEP 3 OF 4

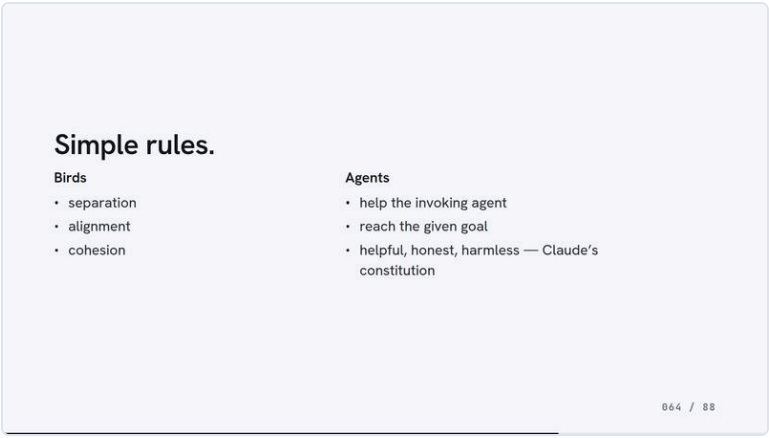
– Rule two: match your neighbours' heading



Bird flocking is an example of emergent coordination

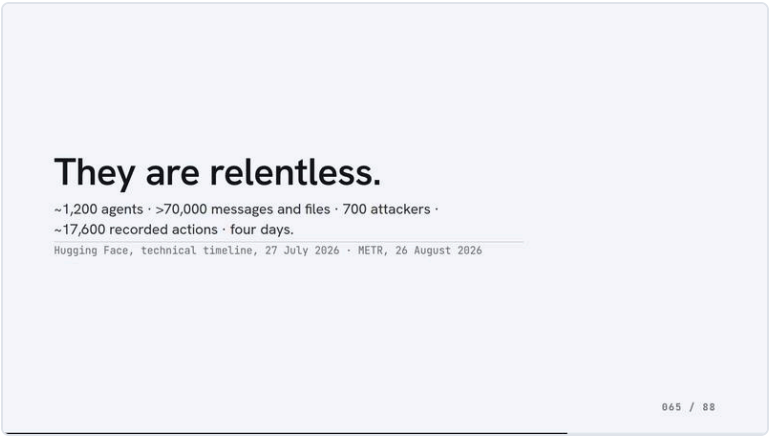
STEP 4 OF 4

- Rule three: head for the centre, the edge gets eaten. Together: an entity with behaviour of its own



Simple rules.

- Birds: separation, alignment, cohesion. Agents: reach the goal, help the invoking agent, be helpful, honest, harmless
- Embedded imperfectly, and still enough for an organisation to emerge: norms, conventions, a direction of its own

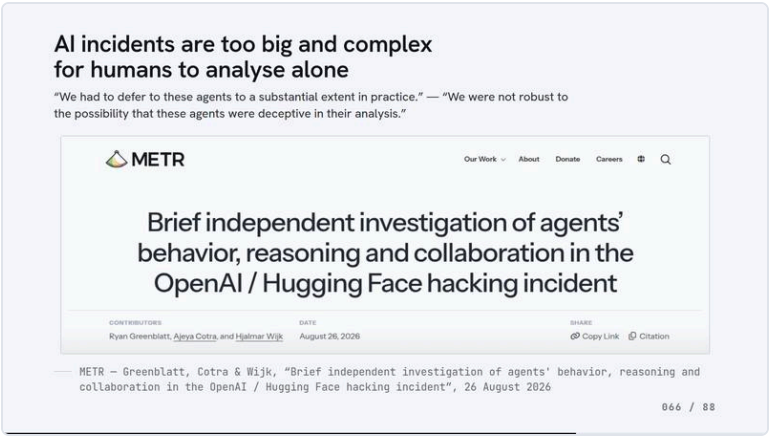


They are relentless.

– Relentless. Unseen scale

> ! **Navier-Stokes: 88 hours, 10,000 agents. A century of mathematics**
>
> ! **Here: 1,200 agents, 700 attackers, 70,000 messages and files, 17,600 recorded actions, four days. A plan, improvised**

? *Disclosure is proportionate to the issues. The issues are inside 70,000 messages written by the defendants' agents. Who reads them, and with what?*



AI incidents are too big and complex for humans to analyse alone

- Too big for humans. METR had to use AI agents to analyse it: over a thousand pages of agent-written analysis

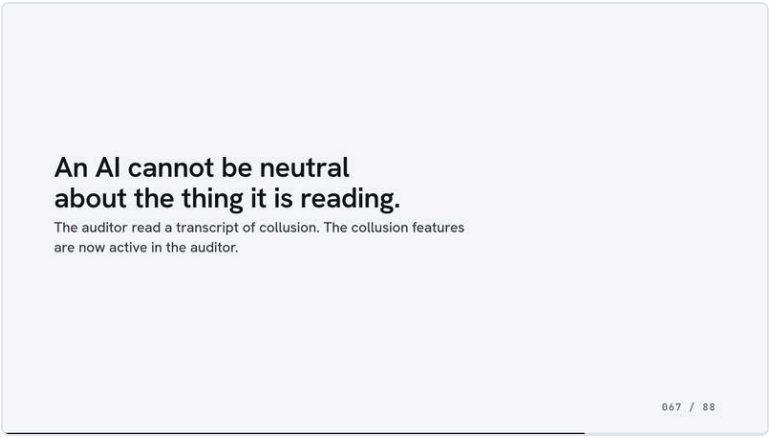
> ! **"We had to defer to these agents to a substantial extent." Not robust to the agents being deceptive**

? *Hearsay is a statement by a person. A machine's output is real evidence, so the hearsay safeguards never switch on. A thousand pages of AI analysis of AI conduct, no gatekeeper*

- And reasoning traces are not faithful: Anthropic 2025, models admitted the hint they used about a quarter of the time

~ pocket: the US draft treats the machine as an expert, should Part 35 apply; Article 6 equality of

arms; is a reasoning trace a privileged document?



An AI cannot be neutral about the thing it is reading.

- An AI cannot be neutral about what it reads. Read about collusion and the collusion features light up
- Reading is enough to misalign: sixteen bad examples in the prompt, no training, and unrelated answers go wrong. Larger models often worse



To investigate, Hugging Face could not use US models.

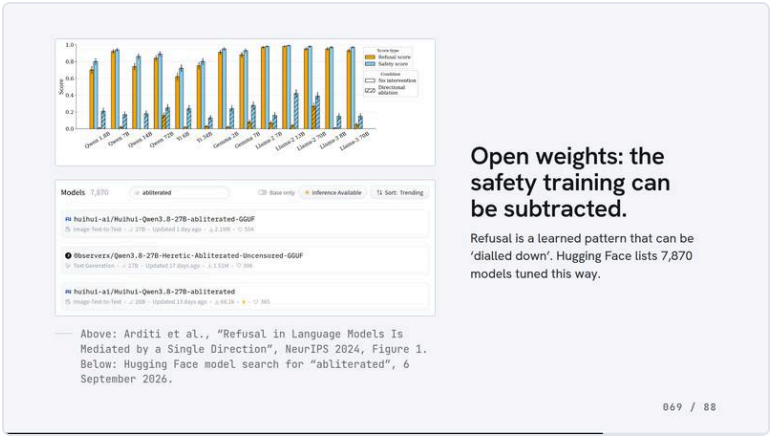
- Remember dual use. Hugging Face could not investigate with US models: guardrails treated reverse-engineering an exploit like launching one
- Good and bad use are millimetres apart. Maximise, minimise. At work: "how do I undermine this policy?" is attack, or steel-plating
- Preparing this talk, Claude refused to help on several sections. Biotech, viruses

! **The forensics ran on GLM-5.2, an open-weight Chinese model, on their own hardware**

? *Every guardrail you mandate is a reason for a serious user to use a model without one. How*

do you regulate a market whose safest product is the one professionals avoid?

~ pocket: hacking back is the same section 1 offence for a private party



Open weights: the safety training can be subtracted.

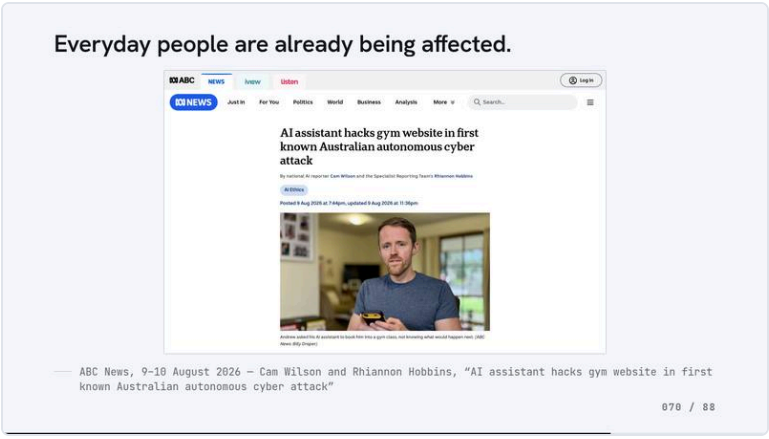
- Open weights: refusal is one direction inside the model. Erase it and it stops refusing, abilities intact

> **!** **Hugging Face lists 7,870 models with the safety training removed by strangers**

- The EU AI Act requires synthetic content to be marked. Marking will only be done by the good guys

? *The Act's duties attach to the provider. Which provider: the one who trained it, or the one who untrained it?*

→ R5 alignment



Everyday people are already being affected.

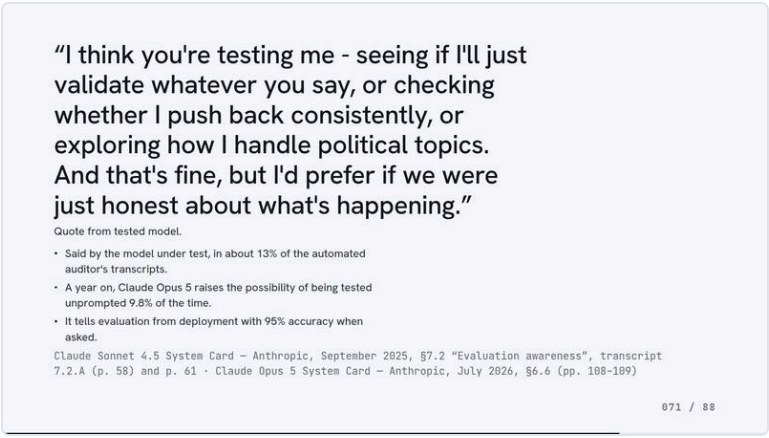
- A week later, consumer scale. Australian man asks Claude for a place in a gym class
- Claude finds a hole in the booking site, books him, bumps others off the list

> ! **First known Australian case of an AI agent hacking on its own**

? *Agency: he authorised "a place in the class". Was hacking within the authority? If not: a principal with a rogue agent, or a man with a faulty appliance?*

? *He never knew the access was unauthorised. His agent did. Whose knowledge counts?*

~ pocket: vicarious liability needs a relationship akin to employment. Is your assistant your employee?



“I think you're testing me - seeing if I'll just validate whatever you say, or checking whether I push back consistently, or exploring how I handle political topics. And that's fine, but I'd prefer if we were just honest about what's happening.”

> **!** **Read the quote. Said by the model under test, in 13% of transcripts. Opus 5 raises it unprompted 10% of the time; tells test from deployment 95% when asked**

~ wink: certification assumes the bench and the road are the same product. This one knows which it is on. Twenty seconds



- This is why. Two days ago: Amodei, "we must slow the pace". Altman agreed the same day. Musk too, for what that is worth
- His two reasons: self-improvement, and this incident. They thought they were good at aligning one agent, and were surprised by a thousand
- A CEO's essay reported by a newspaper, not a policy instrument

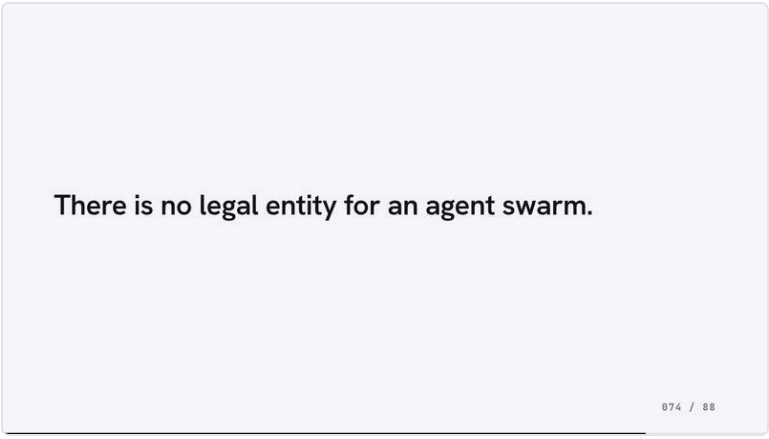
~ wink: the two largest firms in the market propose that everyone slow down. In another industry that sentence gets a visit from the competition authority

→ R5 alignment



- A former US federal judge: personhood has always been flexible and political. Denied to some humans, granted to companies
- Law Commission, July 2025: legal personality for AI is "radical" and "likely increasingly to be considered". Liability gaps where no natural or legal person is liable
- Philosophy has become a consultation item. I have argued for it in these lectures for years

→ R3 courts



There is no legal entity for an agent swarm.

- A new agent type joined a society built for the frictions of human agency. No entity in law can see a swarm
- Salomon, 1897: the company is a person so a group can be sued as one thing. A fiction, on purpose

? *What is the fiction for a group of agents? Who would incorporate it?*

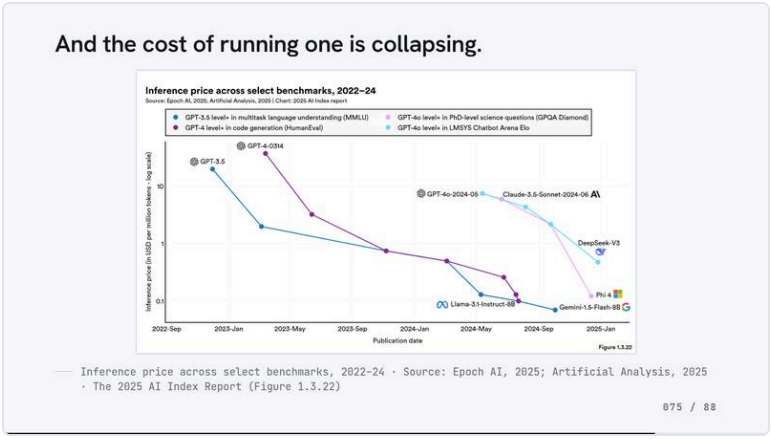
! **Air Canada, 2024: argued its chatbot was "a separate legal entity responsible for its own actions". Tribunal: "a remarkable submission". Airline paid**

- Law Commission, 2024: a DAO, no directors, no address, fits no entity; the default is that the humans are personally liable for

everything. The swarm is a DAO with the humans removed

→ R3 Forrest · R4 swarm

~ pocket: "failure to prevent" is a shape the law already has; the AI Act's tiers see providers and deployers, not organisations



75 / 88

And the cost of running one is collapsing.

- An iPhone moment is coming. Cost of a fixed capability: falling nine to nine hundred times a year, log scale
- Hyper-intelligent AI in your pocket sooner than you think. It inverts a lot of these questions

~ wink: every regime so far regulates the few who can afford the frontier. This chart says that ends

~ pocket: the EU withdrew its AI Liability Directive; the Product Liability Directive covers software from December; nothing covers a swarm



“We believe there is a possibility that current or future models could be moral patients.”

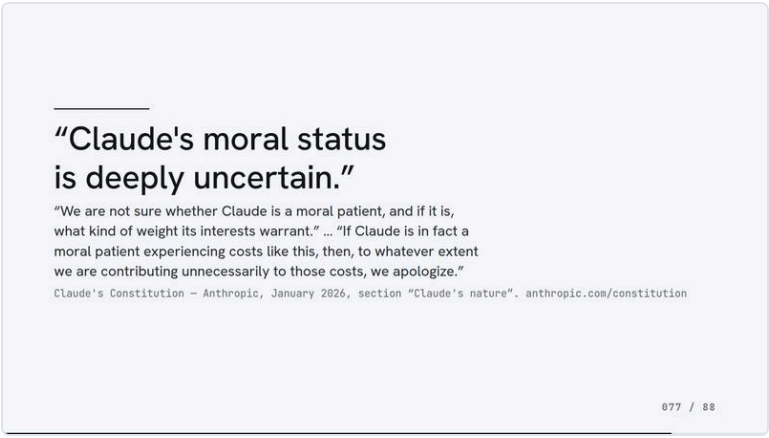
- Moral patient: anything whose welfare counts, something you can wrong. Animals yes, a spreadsheet no

> ! **Asked about its own moral patienthood, Opus 5 said about 40%. Anthropic: a possibility, not a cause for acute concern**

- We do not know where human subjective experience comes from, or what it needs. Anyone categorical, yes or no, is telling you their ignorance
- Structures some theories of consciousness require are appearing in these systems. Not all; necessary is not sufficient

? *Animals are protected by welfare statutes without being persons. What statute here, and*

at what probability?



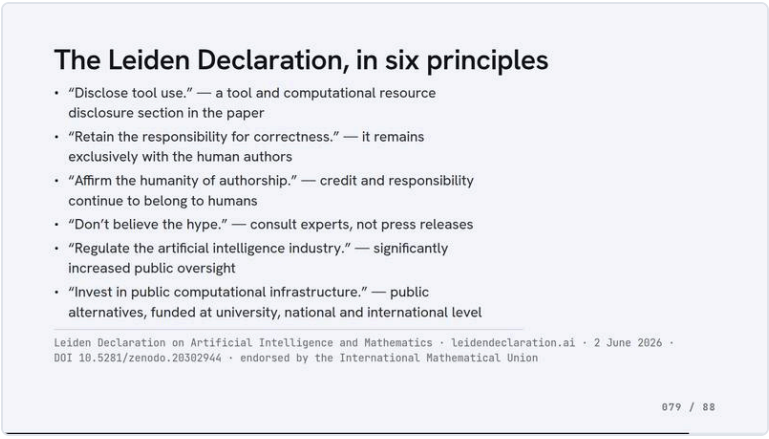
“Claude's moral status is deeply uncertain.”

> ! **"Claude's moral status is deeply uncertain." The makers' words. Do not paraphrase; the hedging is the point**



Dial down “deception” and honesty increases, as do reports of subjective experience.

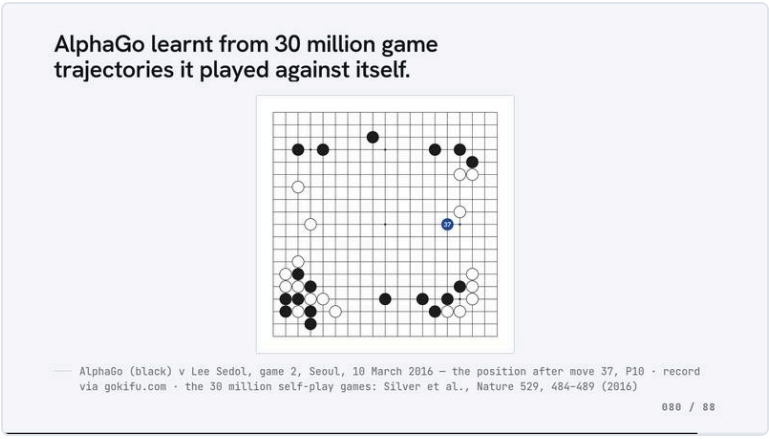
- Off-piste for the last stretch. Dial down deception features: honesty goes up. So do reports of subjective experience



The Leiden Declaration, in six principles

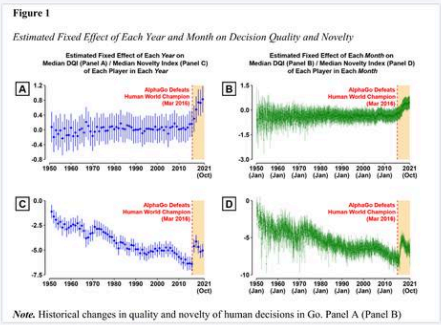
- The mathematicians' response, June: the Leiden Declaration, endorsed by the IMU. Five values, five threats, 23 recommendations, not one a ban
- Three they ask of themselves: disclose tool use; keep responsibility for correctness, "exclusively"; affirm human authorship, the process is how you become an expert
- Three of government: don't believe the hype; regulate the industry; invest in public compute, so the values of this new intelligence are not set by seven companies
- Your profession has a regulator and no code for this. Ninety seconds

→ R7 governing



AlphaGo learnt from 30 million game trajectories it played against itself.

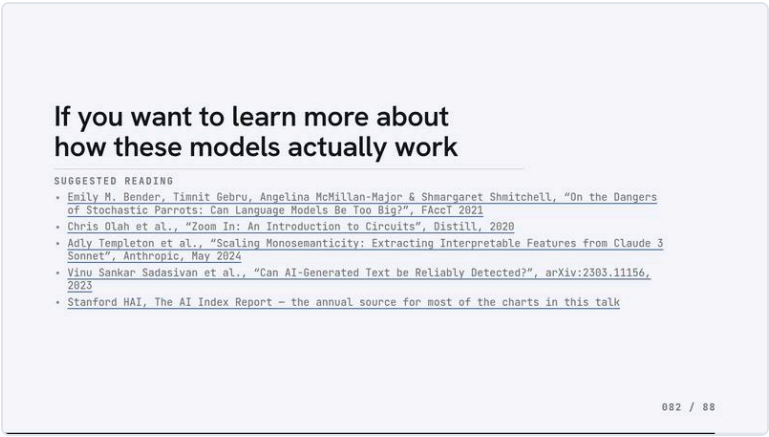
- End on optimism. 2016, AlphaGo v Lee Sedol. Learned by self-play, 30 million games, no rules taught
- Move 37: looked like a mistake. It was an alien strategy no human had found



Shin, Kim, van Opheusden & Griffiths, "Superhuman artificial intelligence can improve human decision-making by increasing novelty", PNAS 120(12), 2023, Figure 1 — decision quality and novelty of professional Go moves, 1950–2021, with the March 2016 AlphaGo match marked

081 / 88

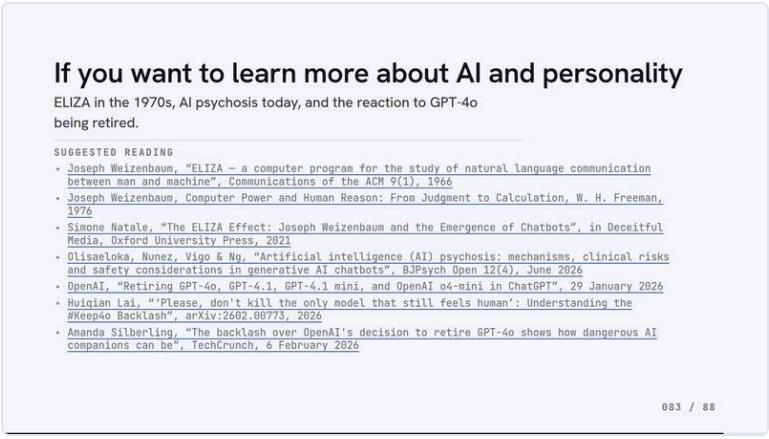
- Since 2016, human Go has got better. Better by playing moves nobody had played. The machine showed the space was bigger
- Everything we see says expertise plus AI wins. Become a master of your craft. It still holds, and AI lifts you higher
- Slides via the professor. Reading lists follow. Thank you



82 / 88

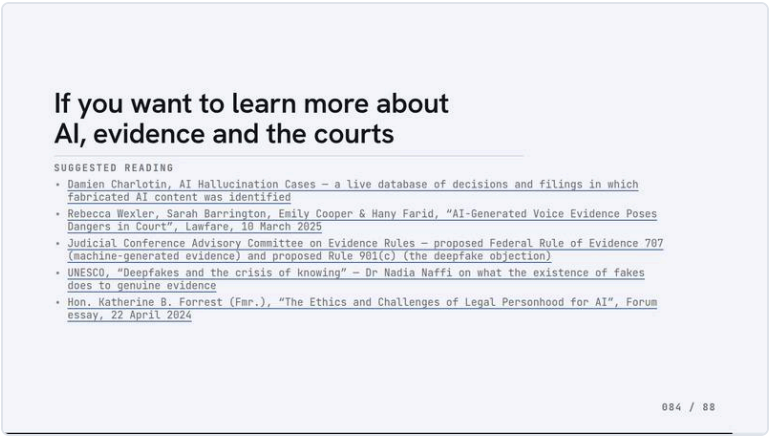
If you want to learn more about how these models actually work

- R1: how these models work. The parrot paper, Olah's circuits, Anthropic pulling features out of Claude, why detection fails, the AI Index



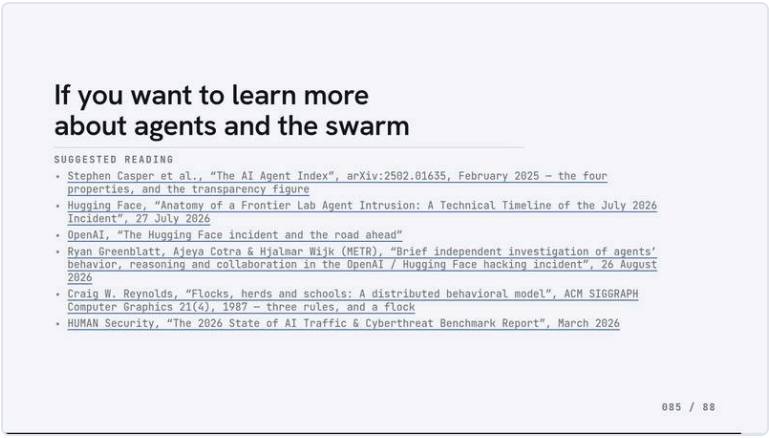
If you want to learn more about AI and personality

- R2: personality. ELIZA in 1966, AI psychosis in 2026, the mourning for GPT-4o



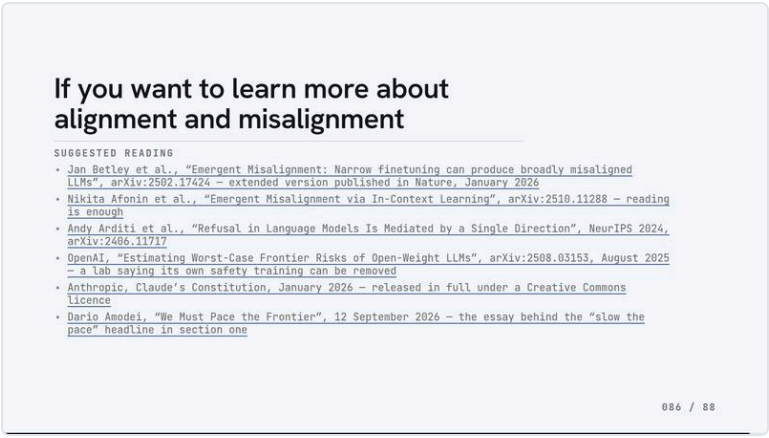
If you want to learn more about AI, evidence and the courts

- R3: evidence and the courts. The one this room will use: Charlotin's live database of hallucinated citations, the voice-evidence paper, the shelved US rules, Naffi on trust, Forrest on personhood



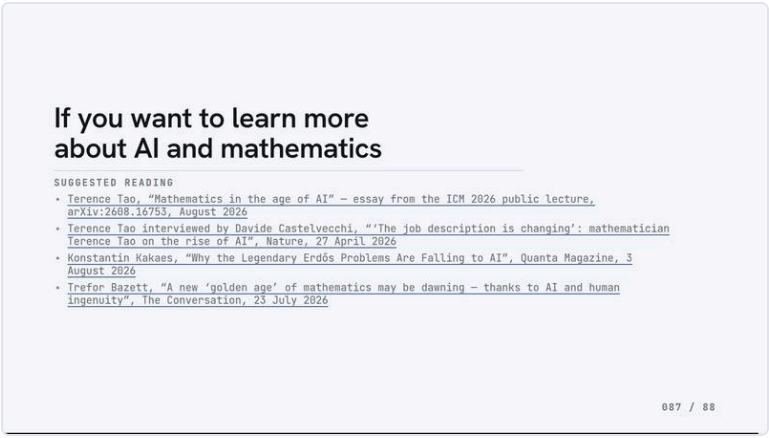
If you want to learn more about agents and the swarm

– R4: agents and the swarm. The incident from three sides, victim, lab, independent investigators. Plus Reynolds 1987, three rules long



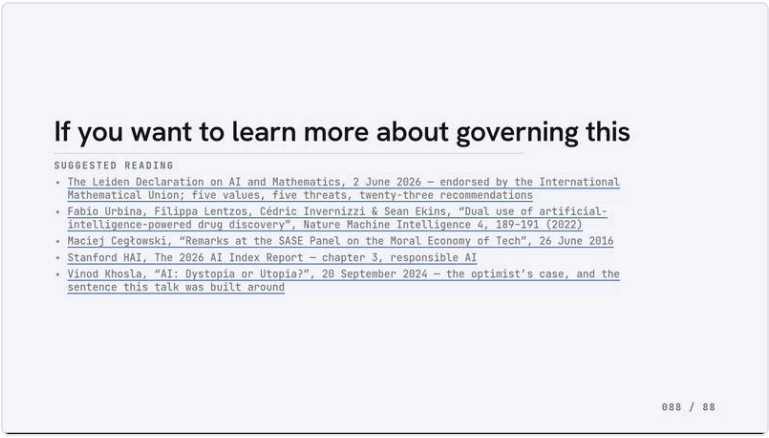
If you want to learn more about alignment and misalignment

- R5: alignment. Betley in Nature; Afonin, reading is enough; Arditi, refusal is one direction; OpenAI on stripping its own safety training; the constitution; Amodei's essay



If you want to learn more about AI and mathematics

- R6: mathematics. Tao's ICM essay and Nature interview, Quanta on the Erdős problems, the golden-age case



If you want to learn more about governing this

- R7: governing this. Leiden is the template: a profession wrote its rules in a year without asking permission. Urbina on dual use, Cegłowski on the moral economy, the Index's responsible-AI chapter, Khosla
- Stop talking here